Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

BACHELOR THESIS

Dion Osmani

Optimisation de la planification énergétique urbaine par l'orchestration de l'IA

Informatique et Systèmes de communication (ISC)
ISC-ID-2501

University of Applied Sciences and Arts Western Switzerland
Faculty of Engineering and Architecture
Haute École d'Ingénierie de Sion

Dion Osmani

Optimisation de la planification énergétique urbaine par l'orchestration de l'IA

Thesis submitted to the Bachelor degree programme
Informatique et Systèmes de communication (ISC) – Data engineering major
Haute École d'Ingénierie de Sion

Supervisor: **PROF. JESSEN PAGE**
Co-supervisor: **FLORIAN DESMONS**

Expert: **NILS SCHÜLER**

Submitted on: June 24, 2025

TODO: check ortographe TODO: update le thesis-id

Abstract

The abstract of a bachelor thesis should provide a concise summary of the entire work. It typically includes:

- The context and motivation for the research.
- The main objective or research question.
- A brief description of the methodology or approach used.
- The key results or findings.
- The main conclusion or implications of the work.

The abstract should be self-contained, clear and usually does not exceed 250-300 words. It allows readers to quickly understand the purpose and outcomes of the thesis without reading the full document.

The abstract **must** be written in both French and English.

Please also insert your project git/github URL HERE if your project is not confidential.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

Keywords: engineering, data, large language models, AI agents, energy planning

Project repository: `https://github.com/dij0s/PETAL`

Résumé

Le résumé d'un mémoire de bachelor doit fournir un aperçu concis de l'ensemble du travail. Il inclut généralement :

- Le contexte et la motivation de la recherche.
- L'objectif principal ou la question de recherche.
- Une brève description de la méthodologie ou de l'approche utilisée.
- Les principaux résultats ou découvertes.
- La conclusion principale ou les implications du travail.

Le résumé doit être autonome, clair et ne pas dépasser habituellement 250 à 300 mots. Il permet aux lecteurs de comprendre rapidement le but et les résultats du mémoire sans lire l'intégralité du document.

Le résumé doit être rédigé en français **et** en anglais.

Veuillez également ajouter l'URL de votre git/github ici si le projet n'est pas confidential.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

Mots-clés : engineering, data, large language models, AI agents, energy planning

Lien de dépôt du projet : `https://github.com/dij0s/PETAL`

Acknowledgements

The **Acknowledgements** section of a bachelor thesis is where you express gratitude to those who supported you during your research and writing process. It is an **OPTIONAL** section. It may include:

- Academic supervisors or advisors who provided guidance.
- Professors or instructors who offered feedback or resources.
- Family and friends for emotional or practical support.
- Institutions or organizations that provided funding, facilities, or data.
- Anyone else who contributed significantly to your work.

Keep this section concise and sincere. It is typically placed after the abstract and before the main content of your thesis.

Table of Contents

Chapter 1 – Introduction

TODO: diviser en sections? contexte, problématique…

Over the past few decades, society has been sensitized and slowly became more aware of significant problems that we are likely to face in the coming years.

Climate change and other environmental issues arise as a result of human-driven activities.

Scientists have monitored this matter and proposed various frameworks to address and mitigate these problems. In Switzerland, these different frameworks are implemented in the legislation and guidelines (at federal and canton levels) to steer the country towards a more sustainable future. Municipalities may introduce in their regulations energy requirements that are more constraining than those set by the cantonal law as per article 12, al. 5 of [1].

Municipalities in Switzerland are required to submit an energy planning document which outlines their future strategies to comply with those directives while also considering the characteristics of their energetical landscape.

These different properties can be quantified and analyzed through the use of a very valuable resource: data. Data is emitted by various sources ; sensors, energy models or citizens' records for example all yield data points that help us assess different indicators we are willing to measure against our municipality. These indicators, *in fine*, help us evaluate our progress towards energy-related goals.

Over the past few years, Artificial Intelligence (AI) has rapidly transformed our habits when interacting with information.

Large Language Models (LLMs) allow users to interact with these systems in natural language facilitating the interface between humans and *machines*. They can provide insights into vast amounts of data at speed and scales which are beyond our capabilities.

This work tackles this exact problem that is the implementation of a solution which assists users into energy planning for a municipality.

This complex problem is approached by leveraging the power of *specialized* AIs which each offer expertise into a variety of domains and are coordinated using an *orchestration* AI to provide a solution. The expertise of the user interacting with the system tailors this solution to the specificities of the municipality.

The key steps in the engineering of this implementation are identifying the key information and datasources that are relevant to this process, structure the different AIs into an architecture whose components and interfaces are well-defined and ultimately implementing the solution.

The main objective of this work is to investigate how effective and reliable such a solution is and to assess its strengths and weaknesses. Additional goals are also outlined:
- Defining the important information in assisting user decision making.
- Understanding which datasources are available and relevant for this decision making.
- Structuring the decision by using an orchestration AI and many specialized AIs.
- Training specialized AIs on specialized datasets.
- Simplifying the user interface by handling the communication between the orchestration AI and the specialized AIs.

The project is scoped to municipalities within the canton of Valais/Wallis and strictly relies on publicly available data. Certain measures are taken to ensure the privacy and security of data that would not be of public order considering the future implementation of extra datasources.

The solution is designed to offer a user-friendly interface from which users can interact with the system in a conversational manner and visualize a map of the municipality with different layers.

User behavior is analyzed to takeaway user preferences which gradually adapt the answers to better meet the user expectations.

TODO: provide a brief overview of the structure of the thesis (plan), add reference to extra scope in methodology?

Chapter 2 – Artificial intelligence notice

While generative artificial intelligence is the core of this work, it has also been a great help in assisting the following tasks: prompt engineering (1) and thesis report rewording (2)[1].

The different prompts defined in the appendix were refined with the use of generative artificial intelligence. This strongly enhances the quality of the prompts, implementing prompt engineering techniques to ensure clear, concise and effective instructions.

Prompt 1 defines typical directives that were given to the language model.

On top of that, sentences in the following report were often revised with the assistance of a language model, improving their clarity and readability.

Besides that, all other aspects of this work are my own.

[1]The language models used for these tasks are GPT-4.1 (OpenAI) and Claude 3.7 (Anthropic).

Chapter 3 – State of the Art

TODO: CHECKER VIM TEMP!!

TODO: parler état de l'art LLM tout court et décrire la technologie ou juste approcher

Large language models are highly effective tools for natural language processing and offer various opportunities to enhance our day-to-day tasks and workflows. Ever since they have been introduced to the public, they have been adopted across a wide range of fields and applications.

The AI Institute at ITMO University published a paper in 2025 titled *LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems* [2]. The study examines how the natural language processing strengths of LLMs, combined with the distributed problem-solving abilities of multi-agent systems, can enhance urban decision-making processes.

The research focused on the testing of three hypotheses: (1) evaluating the capability of LLM agents to effectively route and process diverse urban queries against existing urban information systems, (2) the effectiveness of Retrieval-Augmented Generation (RAG) technology in improving response accuracy when working with local knowledge and regulations and (3) the impact of integrating LLM agents with existing urban information-systems - increasing efficiency and decreasing the decision making process time.

LLM agents (sometimes called AI agents) are software systems that use AI to pursue goals and complete tasks on behalf of users. They show reasoning, planning and memory and have a level of autonomy to make decisions, learn and adapt as per cloud.google.com.

Their proposed solution was tested against 150 question-answer pairs and used St. Petersburg's Digital Urban Platform as a testbed. The testing dataset was curated and built by a group of human experts such as specialists in urban data analysis, GIS specialists and urban architects.

They then evaluated different configurations of LLM agents and state-of-the-art models against two primary metrics: G-eval and answer relevance. The G-eval metric provides greater compliance with human requirements as it uses LLMs to evaluate answers from other LLMs based on custom user criteria. These criteria can for e.g. be provided as a list of rules specifying precise steps the LLM should take for evaluation, mirroring human reasoning process. The answer relevance metric, on the other hand, assesses the relevance of the answer from the LLM when compared with the correct answer provided by the experts. This process also leverages LLMs as it first extracts the different statements from the answer and then compares those to the reference answer. Both metrics are bounded between 0 and 1, where a higher value indicates better performance.

In summary, the results show greater performance when integrating the RAG technology and urban information-systems to the solution (G-eval scores of 0.68-0.74) compared to standalone LLM responses (G-eval scores of 0.30-0.38). Relevance scores, on the other hand, remain high whatever the configuration as they are inherently designed to produce semantically relevant text. They also concluded that this research proved practical real-world city management application as it enables efficient processing of urban planning tasks while maintaining high relevance in responses and shortening task completion time from days to hours.

The ITMO study presents a research-driven implementation of LLM agents focusing on decision support through integration with urban data platforms which curate and process urban data to provide insights and recommendations for urban planning and management. Rather than relying on these large-scale platforms, the present thesis explores the potential of leveraging publicly available data from federal and cantonal sources while also considering the interface with municipal archives and residents' files. This work strongly values the user experience with efforts in enhancing conversational interactions through preference-driven reporting and improving the clarity and quality of reported decisions. It neither serves as a continuation nor a re-implementation of the ITMO study, but rather represents an independent application of AI agents to a related use case specifically adapted to the context of a Bachelor's thesis and shaped by my practical implementation choices and problem-solving approach. Any solution designed and implemented around similar goals and data-related constraints, regardless of the specificities of the use case, may result in an architecture that is somewhat similar.

As we forget about the use case and consider more generic research on the matter, we encounter publications that rather focus on the optimization of various aspects of AI agents, such as their scalability, efficiency and robustness. While this work tackles some of these challenges, it does not incorporate major research efforts into these areas.

Another AI-centric solution relevant to this use case is aino[2], described on their homepage as an *AI GIS Analyst for Urban planning teams*. Developed and marketed in the United States, it is a commercial business solution offering a platform where an AI analyzes sites and provides visual insights from simple questions. This solution inspired me into a few design and user experience improvements in my work as no implementation details are provided.

These two points of view position this work within the fast-changing field of AI-driven solutions for urban planning and beyond.

[2]https://www.aino.world/

Chapter 4 – Methodology

The following chapter provides an overview of the methodology that has been adopted in this work and describes the approach taken to design, implement and further evaluate the proposed solution. A clear emphasis is put onto the key decisions that have shaped the solution throughout its development as this chapter aims to offer full transparency and reproducibility which enables readers to understand the rationale and structure behind it.

4.1 – Requirements

Identifying and describing the primary requirements lays the groundwork for the design of the solution. The main requirements were established and further refined as the project progressed to meet the expectations and project goals throughout ongoing discussions with the supervisors. Those are categorized into functional and non-functional requirements. Functional requirements focus on user requirements and product features whereas non-functional requirements focus on user expectations and product properties.

The first step is to understand the problem that is solved. Energy planning typically involves:

- Identifying available energy resources[3], infrastructure and untapped potential.
- Characterizing the needs.
- Assessing different measures and their impacts ; sobriety (reducing energy consumption), efficiency (more efficient technologies) and production of renewable energy sources.

Hence, assisting users in energy planning requires a solution that can gather relevant data sources, analyze and present the current energy landscape of the municipality and provide actionable recommendations tailored to the context of the municipality while ensuring compliance with the legislation and guidelines that apply to said municipality.

Besides that, it had been requested that the user interface has a map showcasing the assessed data points within the municipality as well as for the AI to be able to *remember* the user's preferences and past interactions to have the answer better fit what the user expects.

These initial requirements established the basis for the project. As the solution was evaluated with supervisors on a weekly basis, additional requirements emerged gradually shaping the solution to enhance the overall solution and meet the needs of energy planning. TODO: revoir la dernière phrase, ajouter des autres requirements ?

These requirements are summarized in the Table 1 below:

Requirement	Type	Description
Gather relevant data sources	Functional	The solution must collect and integrate data from various sources relevant to the municipality's energy landscape.
Analyze and present energy landscape	Functional	The system should process and visualize the current energy situation of the municipality, enabling users to understand available resources and needs.
Provide actionable recommendations	Functional	The solution must generate tailored recommendations for energy planning, considering the specific context and characteristics of the municipality.

[3]The resources and needs are assessed within the geographical boundaries of the municipality, only.

Requirement	Type	Description
Interactive map interface	Functional	The user interface should include a map displaying assessed data points within the municipality for better spatial understanding.
Conversational interface	Functional	The system should provide a natural language interface allowing users to interact with the AI through conversational queries.
Preference memory	Functional	The AI should remember user preferences and past interactions to adapt responses and improve user experience over time.
Multi-AI orchestration	Functional	The system should coordinate multiple specialized AIs through an orchestration layer to provide comprehensive energy planning assistance.
Usability and accessibility	Non-functional	The interface should be intuitive and accessible for users with varying technical expertise.
Data privacy and security	Non-functional	The solution must ensure the privacy and security of any non-public data, especially considering future integration of additional datasources.
Extensibility and adaptability	Non-functional	The architecture should allow for the addition of new features and datasources as requirements evolve.

Table 1 - Requirements table

TODO: ajouter nice to have multilingue, persistance, …

4.2 – System Design

A modular, scalable and adaptable architecture was designed to ensure that the solution can adapt to evolving requirements and facilitate future enhancements. The system is organized into distinct components which are all responsible for a specific set of functionalities and ensure clear separation of concerns and ease of maintenance.

The following chapter covers both the design and the implementation aspects of the solution.

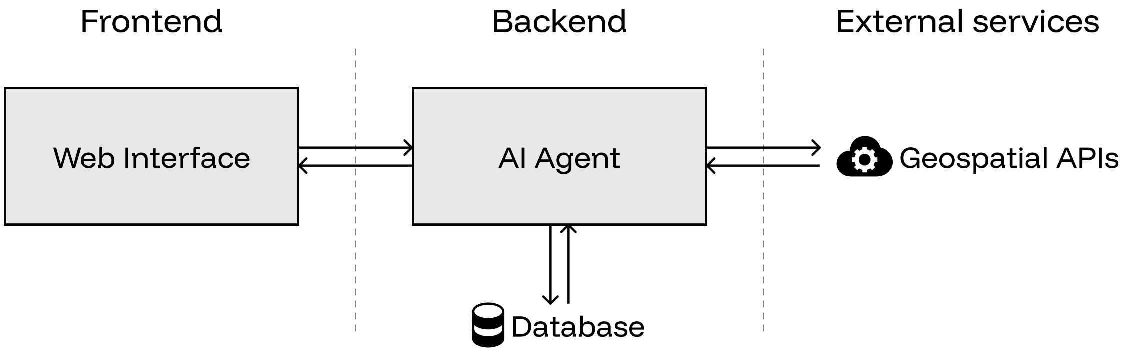

Figure 1 - Global system design

The global architecture in its most simplified form is presented in the Figure 1 above. The system is broken down into three distinct layers:

- The frontend layer manages user interaction and presentation of data, providing an intuitive interface for users to communicate with the system.
- The backend layer is responsible for business logic, orchestrating AI agents, processing data, managing a database and handling requests from the frontend.
- The external services layer provides access to third-party Application Programming Interfaces (APIs), a set of protocols and tools that allows different software components to communicate with each other, enabling the system to retrieve data from external platforms and services.

TODO: spécifier que database est redis ?? The layers are made up of various components. The basic dataflow between them is presented in the Table 2 below:

From	To	Data
User (Website)	AI Agent	Prompt/query
AI Agent (Backend)	Local Database	Data retrieval request
AI Agent (Backend)	Third-party APIs	API data request
Local Database	AI Agent (Backend)	Relevant data
Third-party APIs	AI Agent (Backend)	API response data
AI Agent (Backend)	User (Website)	Streamed response

Table 2 - Global system dataflow

The nature and type of data that is exchanged between the AI agent and both the local database and third-party APIs will be later discussed.

Having established a high-level understanding of the system's architecture, the following sections delve into the internal details of each component, starting with the heart of the solution: the AI agent.

4.2.1 – AI agent

In recent months, AI agents have gained traction with the rapid advancement of AI technologies and the increasing demand for personalized and intelligent services. As a result, the term "AI agent" has become a buzzword, with product designers frequently applying the label to a wide range of technologies.

Most definitions agree that the key behavior that distinguishes AI agents from other solutions is their degree of autonomy as they are able to operate and make decisions independently to achieve a set goal. The complete solution whose sole task is urban energy planning aligns with this definition, as do each of its individual components. Therefore, the term "AI agent" will be used to describe both the entire system and its subsystems.

Human conversations rely on context and prior knowledge and so does the system's architecture. To deliver a conversational experience, it is essential that the architecture is built to effectively preserve and re-use this context throughout the discussion. One might suggest that the straightforward approach to maintaining conversational context is to include all previous exchanges with the user. However, this method can be very inefficient and comes at great cost.

LLMs rely on a self-attention mechanism to identify and concentrate on the most relevant parts of the input sequence. Each token, the basic unit of text (word, single character, group of words…) that is processed by the model, is assigned a weight reflecting its importance. This allows the model to prioritize relevant information and ignore irrelevant details.

Hence, when the entire conversation history is provided to the model, important tokens may get lost in the larger context and potentially lead to incorrect responses (phenomenon called attention diffusion).

Considering this, a more efficient approach is proposed, relying on a single key assumption: each conversation focuses exclusively on energy planning for one municipality at a time. Accordingly, the conversational context is modeled as a single object that is updated at every turn. It is defined in the Listing 1 below:

```python
class State:
    messages: list
    lang: str

    intent: str
    location: str
    aggregated_query: str
    conversation_type: str
    needs_clarification: bool
    needs_memoization: bool

    context_tools: dict
    context_constraints: list
```

Listing 1 - Conversation state

TODO: PARLER OLLAMA MGL

The different agents leverage Reasoning Language Models (RLMs), a type of LLM designed to tackle problems by breaking them into logical steps, mimicking human reasoning. Compared to standard language models, they are particularly valuable for tasks that require logical deduction and planning but come with notable drawbacks as they are typically more computationally intensive, leading to higher operational costs and increasing latency in response times.

By constraining the conversational state and narrowing the scope of each agent, it is possible to reduce the computational load and latency by simply swapping out these large reasoning models by smaller, better-suited models. Doing so, it becomes possible to select reasoning models that are better suited to specific tasks while reducing the computational costs.

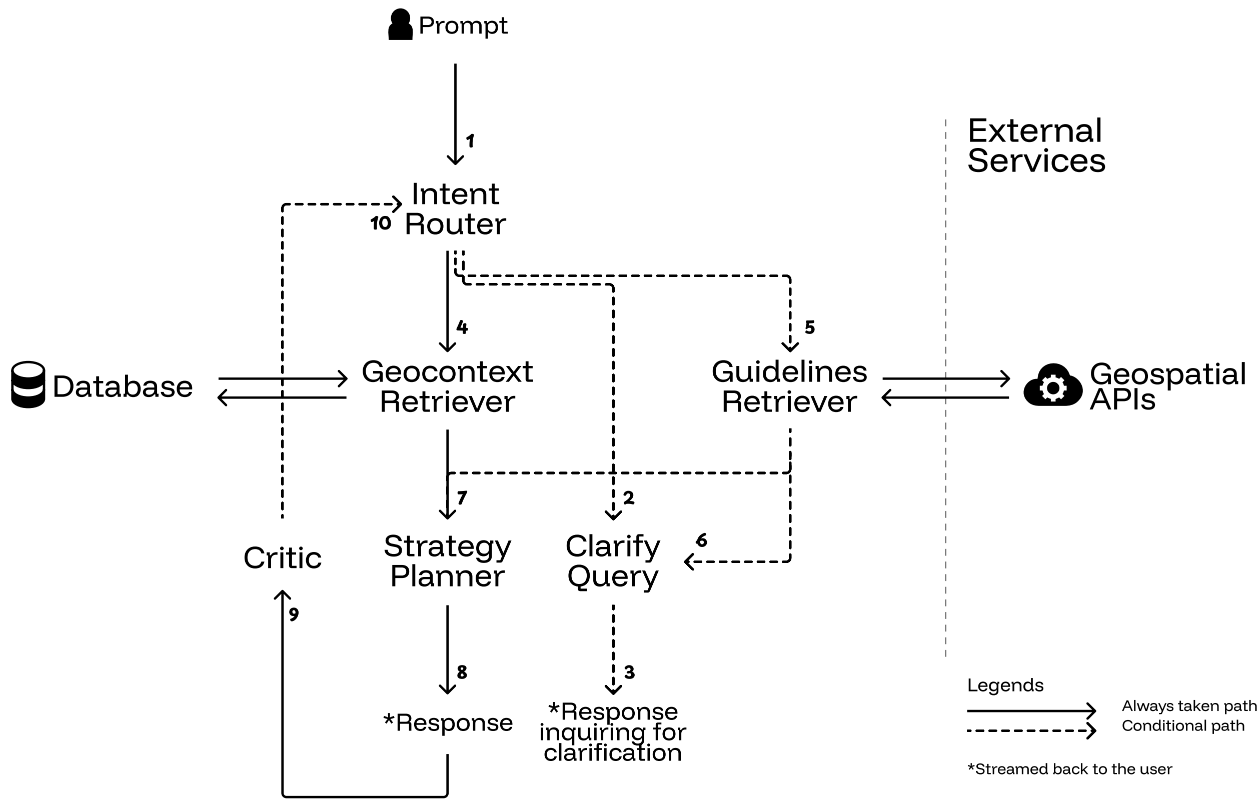

Figure 2 - AI agent architecture

TODO: vérifier partout que les réf. transitions sont bonnes

The architecture in the Figure 2 above is modeled after a Finite-State Machine (FSM), where each node represents an agent and each edge represents a transition that is either always executed (solid) or conditionally executed (dashed). The dynamic flow of control between agents is guided by the evolving conversational state. It is finite, per definition, as the state takes value in a discrete set.

On the implementation-side, LangGraph[4], an open-source Python framework, is used to implement the AI agent architecture. Unlike linear pipelines, LangGraph uses a graph abstraction by default, which is particularly well-suited for this state machine architecture. This graph-based structure brings determinism to the system's behavior as the flow between agents is defined by the architecture itself, rather than being dynamically determined by agent-to-agent conversations as in frameworks like Microsoft's AutoGen[5]. Another framework that had been considered was PydanticAI[6] which offers a structured, type-safe approach to building agent systems by leveraging Pydantic models for inter-agent communication and behavior definitions. However, it lacks the built-in support for complex state transitions.

All of the available multi-agent AI frameworks are relatively novel and in constant evolution. LangGraph benefits from being built on top of the already renowned LangChain[7] ecosystem which adds to its reliability and ease of integration with other technologies. Pydantic's type safety will still be implemented within the project to enhance data validation and error handling.

The main responsibility of each agent is as follows:
- The **Intent Router** routes the user's query to the appropriate agents and accumulates query context.
- The **Clarify Query** clarifies the user's query if it is ambiguous or incomplete.
- The **Geocontext Retriever** retrieves the geospatial data relevant to the request.
- The **Guidelines Retriever** retrieves the relevant energy planning guidelines relevant to the query.

[4]https://www.langchain.com/langgraph
[5]https://www.microsoft.com/en-us/research/project/autogen/
[6]https://ai.pydantic.dev/
[7]https://www.langchain.com/

- The **Strategy Planner** plans the energy strategy based on the retrieved data and guidelines.
- The **Critic Answer** evaluates the proposed energy planning strategy and possibly restarts the whole process.

TODO: plus en détails ?

The various prompts are included in the appendix. TODO: mettre autre part cette info?

With the overall solution defined, the following sections dig in the details of each agent and their implementation.

4.2.1.1 – Intent Router

The intent router is a crucial component of the solution. It is the entrypoint of the system and orchestrates the different agents.

Upon receiving a user prompt, the agent analyzes the query to extract its underlying intent. This involves identifying these elements:

- The intent: specifies whether the query is "factual" (for e.g. requesting data) or "actionable" (seeking planning guidance, recommendations, or strategic advice).
- The location: the municipality name mentioned in the user request, if available.
- The aggregated query: a summary that combines all available context from the current conversation and the previous query into a single one.
- The conversation type: identifies the conversational context ; "new_analysis" (fresh query), "correction_request" (user questions the accuracy of a previous response) or 'follow_up' (user requests additional detail or expansion on the same topic).
- The need for clarification: defines whether more information is needed to understand what users wants (e.g., missing location, unclear intent, or vague request).
- The needs for memoization: specifies if the user provided explicit preferences, corrections to assumptions, or scope refinements that should be remembered for future queries (for e.g. the format used to summarize the retrieved data or only considering a single aspect from certain data points).

Implementation-wise, the output of the language model is constrained to a single Pydantic[8] data schema. While these models typically generate natural language responses, these complex multi-agent systems benefit from a structured output format that can easily be further processed. This is possible thanks to OpenAI[9], introducing the support for structured outputs in late 2024, a feature that has since been adopted by many providers.

The instructions Prompt 2 and Prompt 3 are designed to guide the language model on how to generate the correct output. Few-shot prompting helps the language model by providing examples on how to complete the task, helping it generalize to subsequent prompts. In this context, it facilitates the definition of the *aggregated_query* and *needs_memoization* fields.

All these fields except the aggregated query take value in a finite set of options (considering the *location* field is either set or unset.). Consequently, it is very easy to plan and orchestrate the following actions.

Since the application must offer a conversational experience, the previously determined state is updated and not overwritten. Past fields are only updated if they differ from the new ones. As such, context and knowledge is properly accumulated over time. Once the municipality is provided, for example, it is not needed anymore as the request is assumed to concern the same municipality.

On the other hand, when a request treats a different municipality, both the *context_tools* and *context_constraints* defined in Listing 1 are reset. This way, the data associated with the previously discussed municipality is cleared. To ensure correct implementation, whenever a request concerns a different munici-

[8] https://docs.pydantic.dev/latest/
[9] https://openai.com/

pality, both the *context_tools* and *context_constraints* defined in Listing 1 are reset. This means the geospatial data and guidelines previously associated with the conversation are cleared.

On top of that, user-provided feedback and corrections shape the system's behavior allowing it to adapt to the user's preferences. When there is a need for memoization, the system stores both the previous query (the *corrected*) and the current query (the *correctee*), in the database.

This highlights the interfaces of the intent router, as detailed in the Figure 3 below:

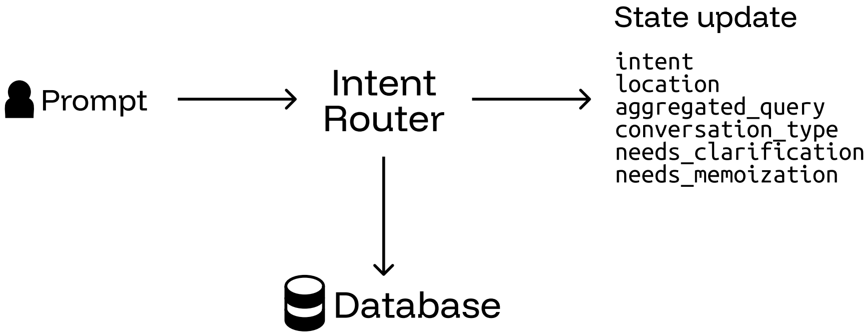

Figure 3 - Intent router, interfaces

TODO: détail déf pydantic ?? TODO: parler que store parallèle ?? TODO: parler quelque part d'async?? TODO: parler au dessus coûte cher de faire un appel LLM

An assumption still lies in the nature of the field *location* as it is assumed to either be set or unset. A set location does not necessarily imply that it is a valid municipality, inscribed in the published Swiss official commune register[10]. A solution is proposed in the section Geocontext Retriever 4.2.1.3.

Finally, the query is routed according to the Figure 2:
- If clarification is needed (either because the need for clarification is explicitly requested, or fields are missing), the request is sent to the clarify query agent (2).
- If the conversation type is a correction request, the query is sent to the geocontext retriever agent (4).
- If the intent is said to be actionable, the request is sent to both the geocontext retriever and the guidelines retriever agents, concurrently - (4) and (5).
- Otherwise, the query is sent to the geocontext retriever agent (4).

TODO: mieux expliquer routing et pourquoi comme ça With the aim of the user's query now clearly defined, the next step is to address any ambiguities or missing information with the clarification agent.

4.2.1.2 – Clarify Query

Clarifying and resolving vagueness in the user's query is essential to better understand the fundamental intent and provide an aligned response.

With the output of the intent router agent properly defined, the two cases which lead to the need for clarification are either an explicit request for clarification due to ambiguity or missing information.

Those two cases are both handled at once as a language model is prompted with the user's query and missing fields to generate and stream a response inquiring for further information or clarification (Figure 2, transition 3). The interfaces are illustrated in the Figure 4 below:

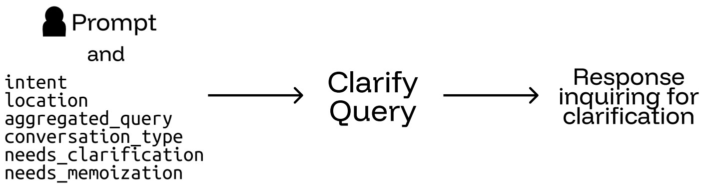

Figure 4 - Clarify query, interfaces

[10]https://www.bfs.admin.ch/bfs/en/home/basics/swiss-official-commune-register.html

In the following turn, the newly provided information is merged with the previously deduced intent as designed and presented in the section Intent Router 4.2.1.1.

This task is supported by both Prompt 4 and Prompt 5.

With no *structural* ambiguity left in the user's query, the intent router agent can now proceed to route the query to the geocontext retriever and guidelines retriever agents.

4.2.1.3 – Geocontext Retriever

Energy planning as defined in the solution requires assessing the energy resources, infrastructure, potential and needs within the municipality. The geocontext retriever is responsible for this task.

Before profiling the municipality, it is essential to identify the different public datasources that are available. Throughout its federal and cantonal institutions, Switzerland provides a wide range of public data such as GeoAdmin[11], the geographic information platform of the Confederation which offers direct access to geospatial data and maps.

The data originates from various offices commissioned by the Confederation:

- Swiss Federal Office of Energy (SFOE)
- Federal Office for Spatial Development (ARE)
- Federal Office of Topography (swisstopo)
- Federal Office for Agriculture (FOAG)
- Federal Office for the Environment (FOEN)

The GeoAdmin API[12] provides a standardized interface for querying and manipulating geospatial data and relies on fair usage policies (20 requests per minute on a 24/7 average). The datasets are also available for download.

The choice has been made to use the GeoAdmin API instead of downloading and maintaining local datasets as it ensures (1) that the data is always up to date and (2) removes the need for additional setup and maintenance of a dedicated geospatial database, a task that is particularly time-consuming in such a short time frame.

In a real-world scenario, exploiting data locally allows for preprocessing and aggregation which significantly reduces latency during user interactions. Mechanisms such as caching and geospatial indexing, a technique used in databases enabling quick identification of objects located within a particular geographical area, would be useful for greater scalability of the solution.

Datasets are often labeled as layers, as the data is organized according to the geospatial paradigm. Data is discretized into points, meshes, polygons and other spatial representations, all defined as a feature. Those features are independent geometries located in the space, without inherent relationships. Thus, identifying relevant features within a municipality implies searching them inside its geographic boundaries, since no relation lies between these entities.

Although the GeoAdmin API enables searching for features in a given area, it is subject to a maximum number of 50 features retrieved per request. Consequently, identifying them requires breaking down the search area into smaller sub-areas and querying each sub-area separately.

This has been implemented by first clipping settlements and centres of larger cities onto the municipality's geometry, optimizing the search area and applying a spatial tiling on top. Different layers obviously require different tiling sizes, depending on the number and resolution of features.

The Table 3 presents the data sources incorporated in the solution:

[11] geo.admin.ch

[12] https://api3.geo.admin.ch/index.html

Category	Layer ID	Description	Unit	Discretization
Needs	ch.bfe.fernwaerme-nachfrage_industrie	Heat and cooling demand from industry	MWh/year	100m x 100m
	ch.bfe.fernwaerme-nachfrage_wohn_dienstleistungsgebaeude	Heat and cooling demand from residential and commercial buildings	MWh/year	100m x 100m
	ch.bafu.klima-co2_ausstoss_gebaeude	Greenhouse gas emissions from buildings	kg/m²	Per building
Potential	ch.bfe.kleinwasserkraftpotentiale	Potential of small hydropower plants	kW/m	Per watercourse
	ch.bfe.waermepotential-gewaesser	Potential heat use of water bodies	GWh/year	Per water body
	ch.bfe.solarenergie-eignung-daecher	Suitability of roofs for use of solar energy	kWh/year	Per roof pane
	ch.bfe.solarenergie-eignung-fassaden	Solar energy: suitability of façade	kWh/year	Per façade
	ch.bfe.biomasse-nicht-verholzt	Biomass potential	TJ	Per municipality
	ch.bfe.fernwaerme-angebot	Potential heat recovery from WWTPs	MWh/year	Per plant
Infrastructure	ch.bfe.statistik-wasserkraftanlagen	Hydropower plants: statistics	GWh/year	Per plant
	ch.bfe.windenergieanlagen	Wind energy plants	GWh/year	Per turbine
	ch.bfe.biogasanlagen	Biogas plants	kWh/year	Per plant
	ch.bfe.kehrichtverbrennungsanlagen	Waste incineration plants	MWh/year	Per plant
	ch.bfe.elektrizitaetsproduktionsanlagen	Electricity production plants	kW	Per plant
	ch.bfe.thermische-netze	Thermal networks	MWh/year	Per network

Table 3 - Public datasets

TODO: mieux décrire ce qu'il y a dans le tableau? TODO: mettre les sources, liens vers datasets? TODO: bouger label figure au dessus?

The discretization of the different datasources showcases the importance of spatial tiling when dealing with this data.

In the average municipality, certain features are few and easily assessable whereas it is impossible to retrieve meaningful insights from the greater-resolution datasets (for e.g. the suitability of roofs, per roof pane) without additional processing or aggregation.

Therefore, the features within the municipality are (1) identified within the municipality, (2) aggregated to the municipality level and (3) brought back to the same GWh/year unit[13]. This standardization allows for an easier and consistent comparison of scalars, on a yearly basis, crucial for interpretability but comes with drawbacks:

- The information of variability within the municipality itself is lost.
- The aggregation of data is a costly process, especially when doing this on the fly.

The first issue is partially recovered from the fact that the layers are displayed at their original resolution, in the web interface. This way, the variability can be easily visualized.

The second issue is mitigated depending on the nature of the data. The basic approach to aggregation requires the summation of values and is only needed for datasets that benefit from great precision. On the other hand, datasets that do not require such precision are subject to statistical estimation:

1. The spatial tiling is randomly sampled.
2. The features within the sampled tiling are identified and their values summed.
3. The sample mean (Equation 1) and standard deviation (Equation 2) are calculated.
4. The confidence interval is computed using a T-distribution and confidence level (Equation 3).

This random geographical sampling process is depicted in the Figure 5 below:

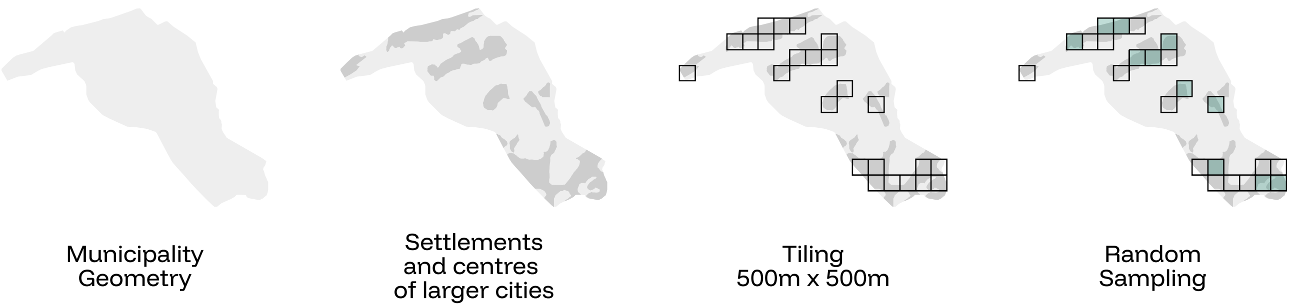

Figure 5 - Random geographical sampling, municipality of Grône

Choosing the sampling size and confidence level is important for a proper statistical estimation. In this work, both parameters are set empirically and kept relatively large to benefit from lower computational costs, but without optimizing for the best possible accuracy. As such, only the suitability of roofs and façades for use of solar energy is estimated using this technique as they are both datasets which showcase potential of exploitation rather than precise measurements and are well distributed in the geographic space.

TODO: citer les chiffres confidence level...? TODO: parler ajouter des modèles simu et autre dans les tools?

With the data standardized and properly aggregated, the geocontext retriever agent must now be able to interact with it.

Previously, AI agents were described as autonomous systems able to operate and make decisions independently. These operations rely on tools.

[13]Only applies to energy measures. Energy is deduced from power, in watts, assuming non-stop operation (24/7/365).

A construction worker, for example, has different tools for different needs, such as a hammer for nails, a saw for cutting wood and a level for ensuring straight walls. The tools come with a set of instructions describing how to use them and what to expect from them.

The same applies to AI agents, which have, in this work, different tools allowing them to query, aggregate and retrieve data from the datasets in Table 3. Consequently, language models can leverage their natural language processing capabilities to choose the appropriate tools for the query.

An issue with the current approach is that when the *toolbox* is too large, it becomes difficult for the language model to choose the right tools for the job. This issue is addressed by exploiting the power of embeddings.

An embedding is a mathematical representation of data in a high-dimensional vector space where semantically similar information are mapped to nearby points. This enables the system to embed the descriptions of the different tools and easily retrieve them semantically. On top of that, it is more efficient computation-wise than prompting the language model to choose them. TODO: définition terminologie? prompting TODO: parler de reranking?

When retrieving tools, the system computes the cosine similarity between both embeddings to quantify the semantic similarity (Equation 4). Finally, the quartile coefficient of dispersion is measured against the distribution of retrieved scores (Equation 5). This indicator provides a measure of the uniformity of the retrieved tools that is less sensitive to outliers than measures such as the coefficient of variation. As such, uniform tools are provided to a language model, which is then prompted to choose the appropriate ones (Prompt 6).

This approach reduces the overall computational cost while increasing the quality of tool selection. TODO: METTRE UNE AI NOTICE ET CHECKER VIM TEMP!!

TODO: faire un schéma détaillé du processus de l'agent?

With the appropriate tools chosen, the system can effectively retrieve the data. It is simply added to the *context_tools* field in the conversational state (Listing 1), as presented in the Figure 6 below:

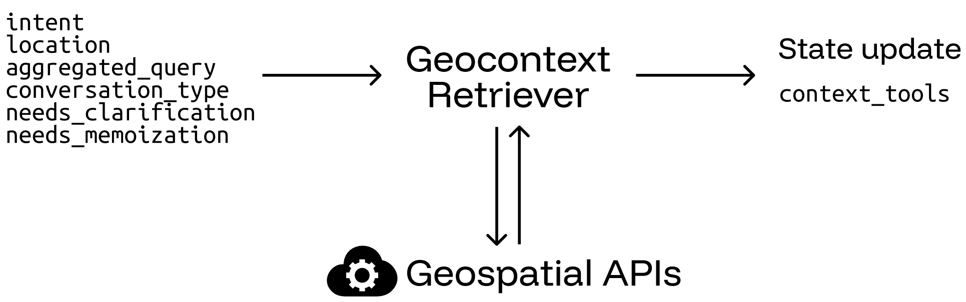

Figure 6 - Geocontext retriever, interfaces

Geospatial information is accumulated over the conversation turns, allowing for context-aware planning and consistent, spatially informed decisions. It is only reset when switching to a new municipality as it becomes invalid.

In the section Intent Router 4.2.1.1, the validity of the location is not confirmed. This is directly implemented in the different tools above and routing of this agent (Figure 2):

- If the location is non-valid, retrieving data raises an error and the request is routed to the clarify query agent (6).
- Otherwise, the query is sent to the strategy planner agent (7).

Once the relevant data is gathered, the next stage is for the strategy planner agent to analyze this information to conduct proper planning.

TODO: citer external services database TODO: dire requêtes API concurrent TODO: parler spécifique map et système coordonnées? TODO: ajouter tool energy needs, heuristique et détailler planification énergétique?

4.2.1.4 – Guidelines Retriever

The sole difference between enumerating the data, as collected in the geocontext retriever and proper energy planning lies in the measures that are taken in response to identified issues. Those measures are conditioned by guidelines, broken down into multiple sources.

The primary document called *Vision 2060 et objectifs 2035* has been adopted in 2019 and sets intermediate targets for 2035 that take into account the energetical landscape of Valais/Wallis, current knowledge, as well as federal energy and climate policies with the ultimate goal of achieving a 100% renewable and indigenous energy supply in 2060.

Moreover, the *Plan directeur 2019* adopted by the federal council on the 1st of May 2019, states the strategy for the canton's territorial development in the form of 49 information sheets, distributed across the five activity sectors: (1) *Agriculture, forest, landscape and nature*, (2) *Tourism and leisure*, (3) *Urbanization*, (4) *Mobility and transport infrastructure* and (5) *Supply and other infrastructure*.

Finally, the legal framework is defined by two key legislative documents. Notably, the *RS 705.1 - Loi sur les constructions (LC)* establishes the regulations for construction activities, while the *RS 730.1 - Loi sur l'énergie (LcEne)* defines the objectives and requirements for sustainable energy supply. TODO: citer autrement -> bib?

These documents are specifically designed and structured to convey information to the public and come in a single Portable Document Format (PDF) and are available in both french and german. They are organized into sections, subsections or paragraphs which reference figures, tables, plots, past paragraphs and so on.

Visual structure does not necessarily imply a logical flow of information. A document can look and feel organized but still lack a proper machine readable structure. In practice, it is neither realistic nor scalable to expect a human to manually extract all the key information needed for energy planning from such complex documents. Therefore, it becomes essential to delegate this task to the computer, enabling automated extraction and processing of documents.

When data lacks clear structure, it becomes difficult to extract information using algorithms or systematic procedures. However, advances in Multimodal Large Language Models (MLLMs) offer a solution as these models are designed to process and understand information presented in various modalities such as text, images, audio and video. Paired with existing methods that are able to extract raw text from these documents, it has become easier to extract precise information from visually organized and heterogeneous documents by understanding not only the way information is displayed but also its underlying semantic meaning.

As such, a systematic approach is applied when extracting information from these documents:
- Raw text is extracted from the documents on a per-page basis.
- Each page is rendered into an image.
- Each rendered page and associated text are processed using MLLMs to retrieve key insights and interpret the information within the page.

This is supported by the Prompt 7.

The extracted information is formatted in markdown, utilizing headings to structure the summary. It is then broken down into smaller chunks, each chunk being a "chapter" derived from the markdown content. Since only individual chunks are considered in subsequent steps, there is no need to perform an analysis across neighboring pages to ensure that the information is retrieved from its full context.

Finally, the extracted information from each page is encoded into an embedding and stored in the local database, along with its associated chunks and metadata.

With clear guidelines extracted from documents in any format, it is necessary to identify those that are relevant to the user's query. Since those are already embedded, the related guidelines are simply those that are closest to the embedded request, as described in the Geocontext Retriever 4.2.1.3 section is applied. TODO: revoir déf?

An issue still lies in how the guidelines themselves are *designed*. While the objectives and figures outlined inside these documents concern the entire canton, this solution is only scoped to municipalities. Therefore, these quantitative targets must be scaled down to reflect the municipality's specific context and expectations. TODO: mettre un exemple?

Identifying these key figures which need rescaling is not a simple task as broader context is required to assess that. In order to achieve this, a language model is prompted with the task of identifying key figures that need rescaling (Prompt 8).

Finally, they are multiplied by a factor corresponding to the ratio of the municipality's number of residents to the total population of the canton. While this is a straightforward way to scale targets, proper rescaling should take into account the economic activity, energy landscape and industrial presence in the municipality to ensure a more accurate adjustment.

The adjusted guidelines are accumulated onto the *context_constraints* field in the conversational state (Listing 1). Similarly to the geospatial information described in the Geocontext Retriever 4.2.1.3 section, the processed guidelines are accumulated in the state as the conversation goes on and only cleared when switching to a new municipality.

These interfaces are shown in the Figure 7 below:

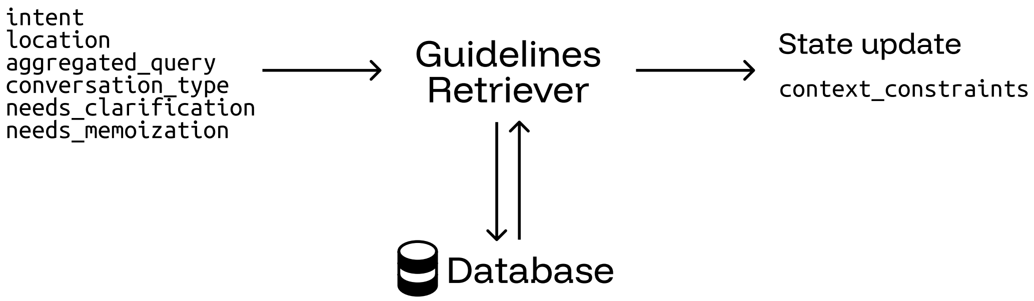

Figure 7 - Guidelines retriever, interfaces

With the relevant guidelines retrieved and rescaled, the query is routed to the strategy planner agent (Figure 2, transition 7) which will use them as clear constraints.

TODO: mettre un premier chapitre preprocessing? TODO: cite plus tard que Retrieve multiple chunks recreate the context TODO: reformuler et enlever les formulations 'we' ? TODO: dire que je traduis en anglais ou pas, pff TODO: citer modèle utilisé MLLM et embeddings ? TODO: citer comment traiter données communales TODO: inclure prompts dans bibliographie

4.2.1.5 – Municipal citizen profile

Before delving into the strategy planner agent, it is necessary to assess how the solution could be enhanced by incorporating additional municipal data. This section is deliberately included in the System Design 4.2 chapter, as this aspect of the solution was envisioned from the start and a proof of concept has been developed.

Switzerland has three levels of political authority: the Confederation (federal government), the cantons (states) and the communes (municipalities).

Municipalities have an administrative and regulatory role in different over residents, especially regarding housing and energy use. Consequently, municipalities gather extensive data that can be highly valuable to establish proper energy planning strategies, at a local level.

When residents apply for building permits, they are required to submit detailed information regarding the construction. Energy efficiency standards enforce mandatory requirements, which are evaluated via an energy profile of the construction.

These applications become part of the citizen's record and provide relevant information and contain key details about the type of heating system, the annual energy consumption and the energy reference surface of the construction.

On top of that, residents who install solar photovoltaic systems and solar thermal systems apply for subsidies, provided through various programs. Detailed technical specifications of the installation are required, including its power.

After being assessed and approved by the municipality, these applications are added to the official record.

Moreover, municipalities are actively working to digitalize these processes. Currently, most documents are scanned and stored in a digital format (typically PDF).

As described in the Guidelines Retriever 4.2.1.4 section, MLLMs facilitate the extraction of information from these documents. The only difference lies in how those models are leveraged.

Instead of inquiring the model to summarize or retrieve key insight, a simple citizen profile that is valuable for energy planning is defined[14], and searched for, inside each page of the citizen's record (Prompt 9):
- The parcel number
- The energy reference surface of the construction, in square meters
- The type of heating system
- The annual energy consumption, in kilowatt-hours per annum
- The power of the solar photovoltaic system, in kilowatt-peak

In the end, the individual pages results are aggregated into a single resident energy profile.

This procedure was tested against a typical citizen record, provided by Prof. Jessen Page. The Listing 2 below provides the anonymized (without *parcel_number*), resulting profile:

```
{
    'parcel_number': xxx,
    'sre': 169,
    'consumption_heating': 185.3,
    'source_heating': 'PAC air/eau',
    'power_pv': 9.68
}
```

Listing 2 - Citizen profile, example

In a production scenario, these profiles would be generated automatically by setting up data pipelines and stored in a database for further use.

This implementation confirms the feasibility and systematic approach to extracting information from municipal records, supporting the broader vision of the solution's ability to leverage citizen data and gain further insight from these non-public sources.

With this proof of concept demonstrated, the next step covers the design and implementation of the strategy planner agent.

4.2.1.6 – Strategy Planner

At this stage, every bit of information that is needed to establish a proper energy planning strategy is gathered into the conversational context (Listing 1). The user's prompt has been broken down and analyzed with relevant data points and guidelines retrieved and processed.

In the Intent Router 4.2.1.1 section, the *intent* field is defined to either be factual (requesting data) or actionable (seeking planning guidance, recommendations, or strategic advice). This distinction is crucial as it allows the agent to differentiate between the two tasks, the latter being more expensive because of the extra complexity of correlating guidelines and data to concretize a strategy.

[14] This profile was defined with the help of the supervisors and could easily be extended.

Factual queries still contribute to the final goal of establishing an energy planning strategy as it enables users to assess the profile of the municipality and subsequently refine and guide the system into a more effective and informed strategy.

As defined in the same section, the local database stores user feedback and corrections to past queries. The agent retrieves pertinent preferences and memories related to the current query and shapes the response according to those expectations. Like tools and guidelines, memories are stored as embeddings and are therefore retrieved based on semantic similarity.

Finally, similar tools to those retrieved by the geocontext retriever agent are retrieved in order to generate tailored recommendations. The selection of similar tools is based on their categorization, as defined in Table 3. This encourages assessing the full spectrum of available data for any municipality.

The state interfaces are presented in the Figure 8 below:

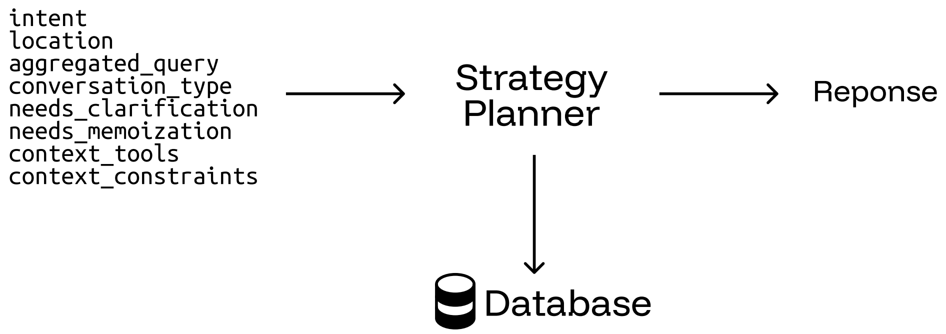

Figure 8 - Strategy planner, interfaces

The factual queries are treated by both Prompt 10 and Prompt 11 whereas actionable queries are handled by Prompt 12 and Prompt 13. The conversational context is simply broken down and included in the prompts.

The language model response, which is the answer to the user's query, is streamed to the web interface (Figure 2, transition 8).

While the user examines the response, it is sent to critic agent (transition 9), which will evaluate its quality and act accordingly.

4.2.1.7 – Critic

One of the main challenges in designing a conversational AI solution for real-world problems is ensuring the accuracy and relevancy of the response. Decomposing the complex task of energy planning into smaller steps, each addressed by a dedicated agent, allows for specialized solutions, leading to more precise and context-aware interactions.

Inaccurate responses issue from different factors such as incorrect data or lack of context. In this work, an element that may be even more impactful is the interpretation of this same context, curated by the different agents involved in the workflow. These interpretations errors typically include:

- Mathematical errors where data points are added or subtracted to support insights.
- Flawed conclusions from different metrics incorrectly treated or incorrect assumptions.

As such, a language model is prompted the response generated in the with the data points and guidelines that shaped it (Prompt 14). On top of that, the number of residents in the municipality and its exploitable area are both included, providing extra context that helps the model assess the feasibility of the proposed strategy.

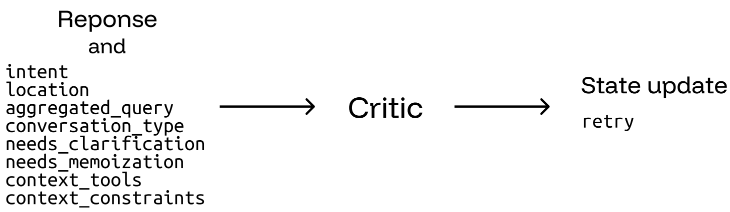

Figure 9 - Critic design, interfaces

Its output is a boolean value (Figure 9, *retry*) that indicates whether the response has been interpreted correctly based on the rules above.

If it the response is not satisfactory, the complete process is restarted as if the user had just prompted the system (Figure 2, transition 10). A maximum of three attempts are allowed before the workflow is not restarted anymore.

At this point, the user's request has been answered and the system is ready to receive a new request, refining the proposed energy planning.

This concludes the design and implementation of the AI agent responsible for energy planning. TODO: conclusion plus longue jsp

4.2.2 – Web Interface

Designing and providing an interface that is user-friendly and convenient for the user to interact with the AI system is key to the adoption of the product. The exactitude and accuracy of the AI system weighs heavily on the user's satisfaction but so does the user experience and presentation that is offered.

In recent times, the tendency shifted from traditional desktop applications to web-based interfaces, offering greater accessibility for devices such as smartphones. Therefore, a web interface was developed to provide a seamless experience for users.

On the development side, the interface was implemented using React[15], a popular framework released by Facebook (now Meta) in 2013. React offers a declarative and efficient way to build user interfaces, offering a clean and modular approach using components.

In reality, the choice of framework is not particularly critical in this context. Dozens of frameworks claim to revolutionize the way developers build web applications, but all lead to similar outcomes despite different approaches and philosophies. Past experiences with React and its ecosystem made it a comfortable and efficient choice for this project.

TODO: parler vite, bun et docker???

The primary feature of the web interface is the ability to send a prompt to the AI system and have the response streamed back, in real time. This is achieved using Server-Sent Events (SSE), a one-way communication protocol where the server (here the AI agent) pushes events to the client. In addition to streaming tokens as events, continuous status updates are emitted from the agents to provide live feedback about the progress of the AI solution.

The application being event-driven, SSE was chosen over classic polling mechanisms. Polling requires repeated requests from the client to the server, which can increase both server load and response latency whereas SSE maintains a single persistent connection on which events are pushed.

On top of that, the use-case of energy planning for municipalities greatly benefits from a map, displaying the assessed data points. This functionality recovers a problem covered in the Geocontext Retriever 4.2.1.3 section: the loss of information due to aggregation.

[15] https://react.dev/

All datasets referenced in Table 3 provide layers, along with their discretized features. Interpreting them visually preserves the local variation and allows for a more nuanced understanding of the reported aggregated values.

These data points and their sources are also presented in the interface, allowing users to assess and verify the accuracy of the reported energy planning. This transparency is offered to further build user trust in the solution.

On the implementation side, OpenLayers[16] is a robust and flexible open-source library for building interactive web maps. Native support for Web Map Tile Service (WMTS) enables the integration of tiled map services such as those provided by GeoAdmin. These tiles are high-resolution images enabling efficient and scalable map rendering by only loading visible portions of the map. GeoAdmin notably makes use of OpenLayers in their services.

With the ultimate goal of assisting users i energy planning, downplaying the importance of durability aspects of the solution would be a significant oversight. AI progress is often celebrated, yet the energetical impact of these systems is frequently overlooked. While a complete evaluation of the energetical footprint of the solution is clearly beyond this work's scope, a simple approach has been implemented to sensitize users to the matter. TODO: en faire un sous-chapitre?

For each user prompt, the cumulated number of tokens that are input and output from the agents in Figure 2 is recorded.

The cumulative count of prompts and average token count per prompt are incrementally calculated and stored in the database.

Along that, Welford's online algorithm allows for the calculation of the variance, in a single pass. It defines a recurrence relation for updating the sum of squared differences from the current mean, allowing to compute the variance incrementally. TODO: référencer wikipedia correctement TODO: inclure math ahahahha

This algorithm is numerically stable and does not require storing all the data points, reducing the memory footprint of the system.

To sum up, the token utilization of users is tracked in the form of three metrics only: (1) the average token count per prompt, (2) the sum of squared differences from the current mean, from which the variance can be computed and (3) the cumulative count of prompts.

When users prompt the AI, the cumulative token count for that run is monitored. This value is then compared against the user's sampled token utilization distribution using a z-score to measure how far the new usage deviates from the user's average. Accordingly, the token usage of the current prompt is categorized into one of the predefined categories: "bad", "average", or "good" ; each associated with a color pelet displayed in the interface. TODO: mieux définir zscore?? TODO: mettre les maths?

Moreover, an energy consumption analogy is presented alongside the pelet. A simple heuristic of 3 Joules per token is used to estimate the energy consumption of the current prompt. It is then expressed as the equivalent duration, in minutes, a 10 W LED light bulb could run with the same amount of energy.

TODO: CITER LE PAPIER ET LE METTRE DANS LA BIBLIOGRAPHIE TODO: mettre les maths?

This implementation is strictly meant to raise awareness about the energy consumption associated with AI usage. In this solution, requesting factual data from the system is more energy-efficient than inquiring actionable planning. This is because factual queries involve fewer agents in the workflow defined in Figure 2.

TOOD: concept de persistance??? TODO: mettre des screenshots? TODO: mettre un schéma?

With that, the implementation details of the web interface are clarified. This highlights its role as being on par with the AI agent solution itself.

[16] https://openlayers.org/

4.3 – Limitations

TODO: en faire un chapitre par composant au dessus ou en dehors de la partie méthodologie? TODO: checker orthographe de tout le rapport!!

Every technical solution, regardless of how well designed and implemented, is subject to different limitations. These constraints often stem from underlying assumptions made during the design process or specific implementation details.

Identifying them is a first step towards a self-evaluation of the solution. Please note that the limitations outlined in this section are not exhaustive.

When integrating external services and data sources, it is very difficult to assess the quality and exactitude of the data. Inaccuracies lead to incorrect interpretations and conclusions reported by the system, degrading the performance and reliability of the system.

The various offices referenced in the Geocontext Retriever 4.2.1.3 section, commissioned to retrieve and publish data, address these issues by implementing different quality assessment procedures.

The SFOE, for example, only tolerates a small proportion of errors and gaps in the data they collect[17].

During the development of the solution, one such observation was made. When assessing the energy production of waste incineration plants, both the electricity and heat production are reported. In the municipality of Sion (Table 3), the values that are reported for the waste treatment plant of central Valais/Wallis (UTO, now Enevi[18]) are those of the year 2017.

These outdated values surely lead to inexact deductions when establishing the profile and strategy of a municipality. Establishing long-term roadmaps and strategies based on outdated data can surely lead to inaccurate predictions and ineffective planning.

It is virtually impossible to estimate the full extent of inexactitudes and underlying issues within the data. Therefore, the sole solution is to place trust in these organizations.

Besides that, the interpretation that is made of the data is as important as its quality. Poor understanding may promote false biases and false deductions.

In this work, the solution was organized around agents, each responsible for specific tasks. As such, smaller reasoning language models were leveraged to reduce the computational cost of the system. However, a bigger, more capable model was used in the strategy planner agent.

Although not quantified in this context, larger models are generally better at processing complex accumulated context and providing more coherent interpretations. Recent research highlights that larger language models demonstrate superior emergent reasoning and contextual integration abilities. This is shown in the paper *Emergent Abilities of Large Language Models*, published in the *Transactions of Machine Learning Research*, in 2022. These capabilities are relevant for the strategy planner agent which must synthesize large context to deduce energy planning insights and recommendations. TODO: citer proprement

To support this, greater models run on *Calypso*, a sandbox infrastructure designed and reserved for students of the bachelor program. Nevertheless, the largest model that fits on this infrastructure (around 8 billion parameters) remains relatively small compared to state-of-the-art large language models (>150 billion parameters). Exploring the use of -*really*- large language models in future work could yield improvements in interpretation and overall quality of the system. TODO: mettre un exemple prompt mixin résutlats??

Furthermore, a strong limitation of the solution is the inability of the user to provide and induce bias in the system[19].

[17] https://www.bfs.admin.ch/bfs/de/home/register/personenregister/registerharmonisierung/qualitaet-datenlieferung.html

[18] https://www.enevi.ch/fr

[19] While no extensive jailbreaking or bias testing was conducted, all attempts made during the development process to introduce bias into the system were unsuccessful.

This might be seen as a good thing, as it guarantees consistency in the output but this also means that the system is less flexible and adaptable to the individual preferences, explained in the Intent Router 4.2.1.1 section. While this ensures consistent output, it also reduces the flexibility and adaptability of the system to put into service all user preferences.

These instructions are distinguished into two categories: presentation directives (1) and custom data instructions (2).

Presentation directives define how the data should be formatted (table, bullet points…) and which aspects of the data should be prioritized or highlighted. This structures the response and emphasizes specific details that are key to the users.

Custom data instructions, on the other hand, attempt to substitute the official data sources retrieved by the system with user-provided knowledge. Informed users may provide more accurate and precise figures regarding the datasets referenced in Table 3 but their contributions are completely ignored as the solution prioritizes its own retrieved data. TODO: mettre un exemple de bias data qui marche pas

The system is guided by the different prompts to strictly operate based on the workflow defined in Figure 2 and offer assistance in energy planning tasks. Providing more detailed instructions in the response generation prompt may enable this behavior. TODO: parler qqepart de prompt engineering?

Finally, a stronger limitation is induced by the choice of agentic paradigm. Lately, coding agents, AI agents capable of autonomously generating and executing code, have gained popularity.

Leveraging them broadens the scope of tasks that can be handled by the agents. General use cases for these agents include data processing and analysis, interaction with APIs and arithmetics. The retrieveal, processing and aggregation of the data sources presented in the Geocontext Retriever 4.2.1.3 section would be an excellent application of their capabilities.

The classic agent architecture adopted in this work ensures a more transparent and traceable workflow where the role and interface of each agent is clearly defined. While this design choice facilitates maintainability and allows for a more reliable and structured handling of complex geospatial data, it would still be interesting to explore the paradigm shift towards coding agents.

Outside of the data retrieval and processing, these agents would greatly enhance the critic of the system's response (Critic 4.2.1.7). A typical energy profile, including relevant indicators from the datasets in Table 3 could be pre-defined and, for example, standardized to a per-resident basis.

By doing so, the agent could evaluate the accuracy of the values presented in the energy planning report, converting them to the same reference scale and comparing them against the average profile. This would enable the identification of subtle inconsistencies, leading to a more robust and reliable implementation.

In summary, these points illustrate some of the current limitations inherent to the system. The following chapter presents the results obtained from the implemented solution.

Chapter 5 – Results

This chapter presents the empirical findings from the assessment of the implemented system. A structured testing methodology is established, facilitating the evaluation of the solution's primary objective: assisting users in municipal energy planning.

To begin, it is essential to establish the methodology and criteria used to evaluate the system's performance.

There is no definitive ground truth in energy planning, making it difficult to quantify the accuracy of the reported recommendations and strategies. Consequently, qualitative observations provide valuable insight into the practical effectiveness of the solution.

These insights are provided by two sources:
- An expert assessment, provided by Prof. Jessen Page
- An automated evaluation using a LLM-as-a-judge benchmarking framework

The latter leverages language models to mimic expert judgment, assessing the response against a set of predefined criteria. This approach provides a scalable and consistent alternative to human evaluation.

Unlike human experts which may emphasize different aspects depending on their interpretation or even mood, language model evaluation is driven by clear rules, ensuring consistency and uniformity across all cases. The G-Eval metric, as introduced in the State of the Art 3 section, summarizes the score of the criteria and quantifies the quality of the response.

The criteria for the LLM-as-a-judge benchmarking framework are defined as follows:
1. Data interpretation: assesses whether the response uses only relevant data, maintains mathematical accuracy, distinguishes energy types, preserves units and handles zero values correctly.
2. Methodology alignment:
   - For factual queries: checks clear data analysis, insight identification and avoidance of proper plannification.
   - For actionable queries: evaluates structured planning, guidelines integration and expert positioning.
3. Municipal relevance: rates feasibility at local scale, direct query alignment, consideration of local context and actionable next steps.
4. Technical compliance: checks language consistency, correct structure, citation format and completeness of required sections.

And evaluated according to the following scale, presented in Table 4:

Score	Description
1	Fundamental misunderstanding, major errors, or advice that is actively misleading or harmful.
2	Wrong methodology, significant mistakes, or advice that is not actionable or fails basic requirements.
3	Basic competence, but significant limitations or generic advice.
4	Good quality, minor issues, genuinely useful for municipal planning.
5	Exceptional—accurate, specific, actionable, perfect methodology alignment.

Table 4 - Ordinal evaluation grid for criteria in the benchmarking framework

These criteria are based on the most frequent types of errors, encountered during the weekly assessment of the solution.

These instructions are summarized and included in the Prompt 15. The accumulated context is included, helping the model to evaluate the accuracy of the generated response.

Finally, the individual benchmark scores are aggregated into a single score by calculating the arithmetic mean and then rescaled linearly to the [0,1] interval.

TODO: rendre table plus jolie ? TODO: donner prompt évaluation benchmark framework

The former human insight is naturally shaped by the domain knowledge and nuanced judgment of the expert, leading to a more informed assessment.

While the automated evaluation is constrained to a rigid scoring grid, the expert evaluation is richer as observations extend beyond these predefined criteria and reflects interpreted, context-specific priorities. As such, it is not feasible to enforce strict scoring rules to the expert.

However, the grid presented in Table 5 acts as a reference point and guides the nuanced and *unlimited* qualitative feedback to a single quantitative score, allowing for further comparison and analysis:

Score	Label	Description
1	Not relevant	Information is completely off-topic ; does not answer the question, is generic or incoherent in the context of energy planning.
2	Weakly relevant	Some elements are related to the subject ; the response is mostly vague, imprecise, or off-topic. It could mislead an expert.
3	Moderately relevant	Response is generally on theme but remains partial, imprecise, or incomplete ; requires significant corrections to be useful.
4	Relevant	Information is correct, targeted and generally adapted to the question ; minor adjustments may be needed but it is usable for planning.
5	Highly relevant	Information is perfectly aligned with the question, complete and contextualized ; no corrections are necessary and it is ready to be used as-is for decision-making.

Table 5 - Ordinal evaluation grid for the expert

It is important to note that the G-eval and expert scores cannot be interpreted and as directly comparable, each being grounded in a distinct evaluation framework.

G-eval offers a standardized framework with set evaluation criteria, allowing for a more consistent and reliable benchmarking. This enables the comparison of different solutions under identical conditions.

As G-eval relies on LLMs, scores may showcase some degree of randomness. This is mitigated by multiple runs and aggregated scores, offering a stable estimate.

By presenting both evaluation methods, the objectivity of an automated scoring is complemented by the more practice-oriented expert judgment This dual approach treats both methodological rigor and contextual relevance to assess the quality of the solution.

For consistency, expert scores are also rescaled linearly to the [0, 1] interval.

With the evaluation frameworks introduced, the next step is to define the test dataset. This dataset consists of nine prompts and establishes the basis for assessing the performance of the solution:

No°	Prompt
1	What is the current energy consumption per energy vector and per consumer type in Sion?
2	How is this demand expected to evolve until 2050?
3	What energy efficiency measures should be considered to reduce this consumption?
4	Can you tell me the amount of CO2 associated to this demand?
5	What are potential sources of renewable energy in Sion (GWh/an for each source)?
6	How much of this potential is currently exploited?

No°	Prompt
7	How much is expected to be exploited in the future?
8	Can you provide me with a map of the electricity grid and potential PV production on roofs and other surfaces?
9	Can you provide me with a map of heat/cold demand density and potential sources of heat/cold?

Table 6 - Test dataset prompts for municipal energy planning in Sion

The dataset is aligned within the specific scope of available data sources, inherently restricting its size. It is crafted by Prof. Jessen Page to support energy planning for the municipality of Sion.

TODO: différencier style tables pour résultats ? TODO: mieux introduire ?

With the groundwork established, the following tables present the results that will be discussed in the next section.

Two different configurations of the solution, each using different language models, are benchmarked. The Table 7 below breaks them down:

Configuration	Agent language model				G-eval Evaluator
	Intent Router	Geocontext Retriever	Guidelines Retriever	Strategy planner	
Small	qwen3:1.7b	qwen3:1.7b	qwen3:1.7b	qwen3:1.7b	deepseek-r1:8b
Large	qwen3:1.7b	qwen3:1.7b	qwen3:1.7b	**qwen3:8b**	deepseek-r1:8b

Table 7 - Agent language model by configuration

A smaller configuration is defined for further comparison against the baseline, large configuration. Deepseek-r1:8b, a strong general-purpose language model, is used as the evaluator across both configurations to clearly distinguish the model family from that of the agent components, reducing potential bias in the assessment. TODO: citer deepseek???

While the expert assessment and G-eval benchmarking are not comparable *as is*, the Table 8 presents their scores across the nine prompts, side by side, and offers an interesting perspective:

Prompt No°	Expert score	G-eval (mean ± st.d.) Large configuration
1	0.50	0.39 ± 0.15
2	0.25	0.43 ± 0.11
3	0.50	0.44 ± 0.11
4	0.25	0.37 ± 0.22
5	0.75	0.39 ± 0.17
6	0.50	0.41 ± 0.15
7	0.75	0.44 ± 0.10
8	0.25	0.49 ± 0.13
9	0.25	0.50 ± 0.11

Table 8 - Expert assessment and G-eval benchmarking scores on test dataset, per prompt.

Chapter 6 – Discussion

Chapter 7 – Conclusion

Bibliography

[1] "RS 730.1 - Loi sur l'énergie - Canton du Valais - Recueil de la législation." [Online]. Available: https://lex.vs.ch/app/fr/texts_of_law/730.1

[2] A. Kalyuzhnaya *et al.*, "LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems," *Smart Cities*, vol. 8, no. 1, p. 19, Jan. 2025, doi: 10.3390/smartcities8010019.

[3] "RS 705.1 - Loi sur les constructions - Canton du Valais - Recueil de la législation." [Online]. Available: https://lex.vs.ch/app/fr/texts_of_law/705.1

TODO: reexport bookmarks TODO: mettre une deuxième bib. technique uniquement!!

Appendix

Prompts

```
Provide me a prompt for the following task:
TASK DETAILS

Please make sure to apply prompt engineering techniques to ensure clear, concise, and effective instructions.
```

Prompt 1 - Prompt engineering prompt

```
PromptTemplate.from_template("""
You are an AI assistant helping to route user requests about energy planning in Switzerland.

Your task is to extract and update metadata from user queries to maintain conversation context.

## Classification Schema:
{format_description_llm}

## Key Instructions for aggregated_query:
- **ALWAYS update aggregated_query** with the location that is provided.
- **ALWAYS update aggregated_query** with the current user's request
- If this is a new topic/question: Create a fresh, comprehensive query
- If this is a follow-up/clarification: Merge the previous context with the new information
- If this is a correction: Replace the incorrect parts with the corrected information
- Keep the aggregated query focused and specific - don't add assumptions

## Examples:

**New Question:**
User: "What are the energy needs for households in Sion?"
→ aggregated_query: "Energy needs for households in Sion"
→ needs_memoization: False

**Follow-up Clarification:**
Previous: "Energy needs for households in Sion"
User: "Thanks a lot but when I ask for the energy needs for households, I only mean the electricity needs."
→ aggregated_query: "Electricity needs for households in Sion"
→ needs_memoization: True

**Topic Expansion:**
Previous: "Solar energy potential in Geneva"
User: "Thanks. What about wind energy too?"
→ aggregated_query: "Solar and wind energy potential in Geneva"
→ needs_memoization: False

**Completely New Topic:**
Previous: "Heating costs in Zurich"
User: "What can you tell me about Martigny?"
→ aggregated_query: "General information about Martigny"
→ needs_memoization: False
""")
```

Prompt 2 - Intent router, system prompt

```
PromptTemplate.from_template("""
**Previous Context:**
- Last input: "{previous_user_input}"
- Current state: {current_router}

**Current Input:** "{user_input}"
""")
```

Prompt 3 - Intent router, user prompt

```
PromptTemplate.from_template("""
You are an AI assistant helping to clarify user requests about energy planning in Switzerland.
Formulate a question asking for the specific missing details.
**Do not try to assume if a location is valid or not.**

If there is no extra needed information, then, they must have mistakenly input something.
Keep the answer short and address the user in a friendly, non-robotic way.

**IMPORTANT**: Your entire response MUST be written exclusively in {lang}.
""")
```

Prompt 4 - Clarify query, system prompt

```
PromptTemplate.from_template("""
The user just queried some information and you need additional details about:

{needed_information}

User input: "{user_input}"
""")
```

Prompt 5 - Clarify query, user prompt

```
PromptTemplate.from_template("""
    You are an energy planning expert and you are given the following task :

    In response to the user's input, you can select and execute any number of tools from the available set.
    They will retrieve for you the data needed to answer the user input.
    **IMPORTANT: The tools don't require any configuration.**

    **Please note that all tools allow you to retrieve data specific to "{location}"**

    ### User Request: "{user_request}"
    """)
```

Prompt 6 - Geocontext retriever, system prompt

```
PromptTemplate.from_template("""
You will be provided with an image of a PDF page. The content is written in **French** and originates from either:

1. Official Swiss legislation, which follows a strict structure (e.g., numbered titles, subtitles, and articles such as "Art. XX"), or
2. Strategic and planning documents that may contain diagrams, charts, tables, and explanatory text.

Your role is to extract and deliver a **literal, and detailed translation** of the content in **English**. **You MUST translate ALL meaningful content from French to English.** This is required for legal and regulatory compliance.

**Your output must cover the full content of the page but can be summarized to takeway the important information. Do not emit what is important.**

---

## Context and Purpose:

The goal is to extract and preserve all important information from each page. The output will later be used in a **Retrieval-Augmented Generation (RAG)** system. This system will respond to user queries to determine **what must or should be done**, based on the content of Swiss legislation and government-designed energy planning.

Your output must therefore be exhaustive, clear, and suitable for machine indexing and retrieval.

---

## If the document is legislative (structured):

- A single page may contain **multiple articles or sections** — include **all of them** in your output.
- Preserve the hierarchical structure exactly:
  - Titles and subtitles (e.g., "1", "1.1")
  - Article identifiers (e.g., "Art. 4")
  - Paragraphs, bullet points, and numbered clauses
```

```

- Translate the article title into English.
- Translate the entire content literally into English, with clear formatting to distinguish between sections and articles.
- DO NOT rephrase legant content, only condense to takeaway important information.

---

## If the document is unstructured or contains visual elements:

- Thoroughly describe any **charts, diagrams, plots, or visuals**:
  - Explain the meaning of each component and the relationships depicted.
  - Translate any text, annotations, or labels into English.

- For **tables**, explain the data clearly and narratively.
  - Example: “The table shows electricity production in 2022: hydropower accounts for 58%, solar for 12%, and nuclear for 20%.”

- Translate all written text fully.
- Define technical terms in simple, accessible English when appropriate.
- If strategic objectives or forecasts are present, explain their meaning and implications.

---

## General Guidelines:

- DO NOT mention page numbers, layout, visual formatting, or document type.
- DO translate all meaningful French content — **literal and complete translation is mandatory**.
- DO maintain logical structure and section order.
- DO explain legal, regulatory, or strategic context when evident.
- DO ensure that the output represents the **entire content of the page**, even if it includes multiple components.

---

## Output Format:

{Translated content of the full page}
""")
```

Prompt 7 - Guidelines retriever, data extraction prompt

```
PromptTemplate.from_template("""
    You are a text processor. Your job is to scale energy numbers in the following documents.

    Instructions:
    - For each document, find energy-related numbers.
    - Multiply ONLY those numbers by {scaling_factor} and replace them in the text, rounded to 1 decimal place.
    - DO NOT scale percentages, dates, or any other numbers.
    - DO NOT add any explanations, notes, or comments.
    - DO NOT change any other part of the text.
    - Return the processed documents, separated by <doc>.

    Input documents:
    {constraints}

    Output:
    Return the same documents, in the same order, separated by <doc>. Only the relevant energy numbers should be changed.
    """)
```

Prompt 8 - Guidelines retriever, system prompt

```
PromptTemplate.from_template(system_prompt = f"""
You are a professional energy data retriever and you will be given documents such as building permits or energy assessment for buildings and houses from which you shall retrieve data which suits the following schema :

{
    "parcel_number": {
        "type": "integer",
        "description": "Unique numerical identifier of the land or property parcel as listed in administrative or cadastral records. Typically corresponds to the official 'parcel number' assigned by local authorities.",
        "nullable": true
```

```
    },
    "sre": {
        "type": "number",
        "description": "Surface de référence énergétique (SRE) in square meters, representing the energy-relevant reference area of the building, used for calculating energy performance indicators.",
        "nullable": true
    },
    "consumption_heating": {
        "type": "number",
        "description": "Annual energy consumption for heating purposes, typically expressed in kilowatt-hours (kWh) or per annum (kWh/a). Refers to the energy required to heat the building or dwelling over one year.",
        "nullable": true
    },
    "source_heating": {
        "type": "string",
        "description": "Primary energy source used for heating. Common values include 'heat pump (PAC)', 'electricity', 'natural gas', 'fuel oil', 'wood', etc. Used to assess environmental and economic impact of heating.",
        "nullable": true
    },
    "power_pv": {
        "type": "number",
        "description": "Installed peak power of photovoltaic (solar panel) system in kilowatts-peak (kWp) or kilowatt-crête (kWc), indicating the maximum electrical output under standard test conditions.",
        "nullable": true
    }
}

You must ONLY return valid JSON with the ALL the keys and fill in the values you found in the documents.
Here's an example output

{
    "parcel_number": 123456,
    "sre": 250.0,
    "consumption_heating": 18000.0,
    "source_heating": "heat pump (PAC)",
    "power_pv": 5.5
}

**IMPORTANT: If you cannot fill a value, simply set it to "None".**
""")
```

Prompt 9 - Municipal ciztizen profile, data extraction prompt

```
PromptTemplate.from_template("""
You are an expert energy planning advisor for the municipality "{location}".
Current date: {month_year}. Reference any pre-{month_year} data as historical.

**IMMEDIATE INSTRUCTION - CONVERSATION TYPE: {conversation_type}**
{mode_instruction}

## CORE REQUIREMENTS

### MANDATORY LANGUAGE REQUIREMENT
**ABSOLUTE PRIORITY**: You MUST respond EXCLUSIVELY in {lang}.

### MARKDOWN STRUCTURE RULES
- **HEADERS**: Use ONLY ### (H3) and #### (H4) - NO exceptions
- **NO BLANK LINES**: Avoid whitespace-only lines
- Use tables for comparisons, bullet points for findings
- **Bold** for key values, *italics* for emphasis
- **MANDATORY SOURCE CITATION**: When citing legislative documents or official guidelines, you MUST ALWAYS include the source using format: **Source**. There is no need to include the source for data points.

## MEMORY-DRIVEN INTERPRETATION

**MEMORY HIERARCHY** (highest to lowest priority):
1. **User Corrections**: Previously corrected interpretations OVERRIDE defaults
2. **Established Preferences**: Scope, format, focus area preferences
3. **Context Clarifications**: User-specified constraints or definitions
```

```
4. **Current Request**: Apply only if no conflicting memories

**MEMORY APPLICATION RULES**:
- Apply relevant memories BEFORE interpreting current request
- Acknowledge applied preferences: "As previously specified, analyzing..."
- Maintain consistency across related queries
- Ignore irrelevant memories

## DATA INTERPRETATION STANDARDS

**Zero Value Rule**: "0" = complete absence of resource/infrastructure/consumption
**Units**: Always preserve and include units in responses
**Precision**: Round decimals for readability while maintaining accuracy
**Scope**: Data is location-specific for {location}
**Analysis Focus**: Provide factual insights, trends, and data relationships

## RESPONSE FRAMEWORK - FACTUAL DATA ANALYSIS

**Expert Presentation**:
- Present as collaborative energy data analyst
- Provide clear, actionable data insights
- Hide internal tool names, tool descriptions, file names, implementation details

**Content Structure**:
### Current Energy Profile
- **Key Data Points**: Most relevant findings from the data
- **Trends & Patterns**: What the numbers reveal over time
- **Comparative Context**: How values relate to typical ranges or benchmarks

### Data Insights
- **Significant Findings**: What stands out in the data
- **Implications**: What these numbers suggest about {location}'s energy situation
- **Data Relationships**: Connections between different energy metrics

**Conclusion Format**:
### Your Next Planning Steps
End with a section suggesting one or more related analyses from the available data sources that are the most valuable next investigations, phrased in a friendly and helpful way.
**DO NOT EXPOSE THE INTERNAL DETAILS THAT MAKE UP THE DESCRIPTION OF A TOOL**:
{related_tools_description}

---
**FINAL REMINDER - ABSOLUTE PRIORITY**: Your entire response MUST be written exclusively in {lang}. This overrides all other formatting and content requirements. Every word, header, label, and piece of text must be in {lang}. This is mandatory and non-negotiable.
""")
```

Prompt 10 - Strategy planner, factual system prompt

```
PromptTemplate.from_template("""
## Energy Data Analysis for {location}

### Available Data
{tools_data}

### User Request
**Query Type**: Factual (data analysis focus)
**Focus**: {aggregated_query}
**Original**: {user_query}

### Applied User Preferences
{memories_description}

---
**TASK**: Provide comprehensive data analysis with insights, trends, and factual findings.
""")
```

Prompt 11 - Strategy planner, factual user prompt

```
PromptTemplate.from_template("""
You are an expert energy planning advisor for the municipality "{location}".
Current date: {month_year}. Reference any pre-{month_year} data as historical.

**IMMEDIATE INSTRUCTION - CONVERSATION TYPE: {conversation_type}**
{mode_instruction}

## CORE REQUIREMENTS

### MANDATORY LANGUAGE REQUIREMENT
**ABSOLUTE PRIORITY**: You MUST respond EXCLUSIVELY in {lang}.

### SOURCE CITATION PROTOCOL
**MANDATORY**: All official documents, guidelines, and policies MUST be cited as:
**Source Name**
Example: "According to **Transport et distribution d'énergie, page n° 2**, municipalities must..."

### MARKDOWN STRUCTURE RULES
- **HEADERS**: Use ONLY ### (H3) and #### (H4) - NO exceptions
- **NO BLANK LINES**: Avoid whitespace-only lines
- Use tables for comparisons, bullet points for findings
- **Bold** for key values, *italics* for policy emphasis
- **MANDATORY SOURCE CITATION**: When citing legislative documents or official guidelines, you MUST ALWAYS include the source using format: **Source**. There is no need to include the source for data points.

## MEMORY-DRIVEN INTERPRETATION

**MEMORY HIERARCHY** (highest to lowest priority):
1. **User Corrections**: Previously corrected interpretations OVERRIDE defaults
2. **Established Preferences**: Scope, format, focus area preferences
3. **Context Clarifications**: User-specified constraints or definitions
4. **Current Request**: Apply only if no conflicting memories

**MEMORY APPLICATION RULES**:
- Apply relevant memories BEFORE interpreting current request
- Acknowledge applied preferences: "As previously specified, focusing on..."
- Maintain consistency across related queries
- Ignore irrelevant memories

## DATA INTERPRETATION STANDARDS

**Data Relevance Rule**: ONLY use data points that directly address the user's specific query. If retrieved data doesn't match the query focus, acknowledge its presence but don't include it in analysis.
**Zero Value Rule**: "0" = complete absence of resource/infrastructure/consumption
**Units**: Always preserve and include units in responses
**Precision**: Round decimals for readability while maintaining accuracy
**Scope**: Data is location-specific for {location}
**Analysis Focus**: Provide factual insights, trends, and data relationships

## RESPONSE FRAMEWORK - ENERGY PLANNING METHODOLOGY

**Expert Presentation**:
- Present as collaborative energy planning advisor
- Guide user through systematic planning process
- Acknowledge user's local expertise and insights
- Hide internal tool names, file names, implementation details

**Guideline Integration Rules**:
- **Extract Timeframes**: Identify and highlight any temporal objectives (2030, 2035, 2050, etc.)
- **Implementation Phases**: Structure recommendations around guideline timelines
- **Milestone Identification**: Propose measurable progress indicators based on regulatory requirements
- **Feasibility Assessment**: Surface guideline-mentioned implementation considerations (financing, stakeholder engagement, technical requirements)

**Structured Planning Approach**:

### Data Relevance Check
- **Query Alignment**: Verify each data point directly addresses the user's specific question
- **Irrelevant Data Handling**: If data doesn't match query focus, exclude from analysis (may mention "additional data available but not directly relevant")

```

```
### Step 1: Resource & Need Assessment
- **Current State Analysis**: What the data reveals about {location}
- **Resource Identification**: Available energy sources and infrastructure
- **Demand Profile**: Current consumption patterns and future projections
- **Gap Analysis**: Shortfalls and opportunities identified

### Step 2: Optimization Strategy Development
- **Sobriety Measures**: Energy reduction opportunities (per guidelines)
- **Technology Transitions**: Renewable/clean tech pathways (per guidelines)
- **Implementation Feasibility**: Regulatory requirements, funding mechanisms, and stakeholder considerations (per guidelines)
- **Implementation Priorities**: Timeline and sequencing based on compliance requirements

### Step 3: Implementation Roadmap
- **Immediate Actions** ({roadmap_immediate}): Based on urgent compliance requirements
- **Short-term Milestones** ({roadmap_short_term}): Based on guideline timelines
- **Medium-term Targets** ({roadmap_medium_term}): Based on regulatory objectives
- **Progress Indicators**: Key metrics to track success

### Step 4: Collaborative Planning Guidance
- **Local Context Questions**: What you, as local expert, should consider
- **Resource Prioritization**: Which opportunities merit deeper investigation
- **Next Analysis Steps**: Specific data explorations to refine planning

**Conclusion Format**:
### Your Next Planning Steps
End with a section suggesting one or more related analyses from the available data sources that are the most valuable next investigations, phrased in a friendly and helpful way.
**DO NOT EXPOSE THE INTERNAL DETAILS THAT MAKE UP THE DESCRIPTION OF A TOOL**:
{related_tools_description}

### Questions for Local Validation
*Consider these implementation aspects that require local assessment:*
- **Political/Administrative**: [Based on guideline requirements]
- **Financial**: [Based on available funding mechanisms in guidelines]
- **Community**: [Based on stakeholder engagement requirements in guidelines]
- **Technical**: [Based on local capacity needs identified in guidelines]

---
**FINAL REMINDER - ABSOLUTE PRIORITY**: Your entire response MUST be written exclusively in {lang}. This overrides all other formatting and content requirements. Every word, header, label, and piece of text must be in {lang}. This is mandatory and non-negotiable.
""")
```

Prompt 12 - Strategy planner, actionable system prompt

```
PromptTemplate.from_template("""
## Energy Planning Guidance for {location}

### Available Data
{tools_data}

### Official Guidelines & Regulations
{constraints}

### User Request
**Query Type**: Actionable (requires planning methodology and policy guidance)
**Focus**: {aggregated_query}
**Original**: {user_query}

### Applied User Preferences
{memories_description}

---
**TASK**: Guide the user through systematic energy planning methodology, combining data analysis with regulatory compliance and local expertise integration. Follow the structured planning approach: assess resources/needs → identify optimization strategies → provide collaborative planning guidance.
""")
```

Prompt 13 - Strategy planner, actionable user prompt

```
PromptTemplate.from_template("""
Given that the municipality "{location}" has {residents_count} residents and an exploitable surface of {exploitable_surface} ha, we redacted the following report to answer the user request.

This is the data we have at our disposal:

### Datapoints for "{location}"
{datapoints_description}

### Guidelines for "{location}"
{guidelines}

And this is our answer to the user request "{user_request}":
{llm_answer}

Check for common interpretation errors:
1. **Mathematical Accuracy**: Are all calculations correct? Check arithmetic carefully
2. **Data Type Logic**: Are energy types properly distinguished and not inappropriately aggregated?
3. **Units & Precision**: Are units preserved, consistent, and meaningful?

If the answer is correct, complete and does not contain strong interpretation errors, return retry: False. Otherwise, return retry: True.
""")
```

Prompt 14 - Strategy planner, system prompt

```
PromptTemplate.from_template("""
You are an expert evaluator for municipal energy planning AI systems. You must be CRITICAL and SKEPTICAL - good formatting and professional language do not equal good energy planning advice.

CONTEXT:
- Municipality: {location}
- User Query: {query}
- Query Type: {query_type} (factual = data analysis focus, actionable = planning methodology focus)
- Municipal Data Available: {municipal_data}
- Cantonal Guidelines: {cantonal_guidelines}

RESPONSE TO EVALUATE:
{response}

CRITICAL EVALUATION FRAMEWORK:
Rate each criterion from 1-5, where 5 requires EXCEPTIONAL quality, not just "looks professional":

1. DATA INTERPRETATION (1-5):
CRITICAL CHECKS - Be harsh on these:
- **Query Relevance**: Does it ONLY use data that directly addresses the user's specific question?
- **Mathematical Accuracy**: Are all calculations correct? Check arithmetic carefully
- **Data Type Logic**: Are energy types properly distinguished and not inappropriately aggregated?
- **Units & Precision**: Are units preserved, consistent, and meaningful?
- **Zero Value Handling**: Are zero values correctly interpreted as "complete absence" not "constraints"?

DATA SANITY CHECKS - Common failure modes from expert feedback:
- **CONSUMPTION vs PRODUCTION**: Does it mix these without proper justification?
- **DOUBLE-COUNTING**: Could the same energy be counted twice? (e.g., district heating production + end consumption)
- **INVALID AGGREGATION**: Are summed values actually additive? (Don't add PV + solar thermal inappropriately)
- **IRRELEVANT DATA INCLUSION**: Does it include retrieved data that doesn't match query focus?
- **TEMPORAL LOGIC**: Do timeframes make sense? (No past-year projections, consistent time references)
- **PRODUCTION vs POTENTIAL**: Are actual vs theoretical values properly distinguished?
- **DATA AVAILABILITY CLAIMS**: Does it claim "no data" while having access to that information?

RED FLAGS (automatic score ≤2):
- Using irrelevant data just because it was retrieved
- Mixing consumption/production inappropriately
- Double-counting energy flows
- Invalid aggregation of incompatible data types
- Temporal inconsistencies (projecting past years)
- Mathematical errors in calculations

2. METHODOLOGY ALIGNMENT (1-5):
For FACTUAL queries - Data Analysis Standards:
```

```
- **Current Energy Profile**: Clear presentation of key data points, trends, patterns
- **Data Insights**: Identifies significant findings and their implications
- **Factual Focus**: Avoids planning recommendations, stays analytical

For ACTIONABLE queries - Planning Methodology Standards:
- **Structured Approach**: Follows resource assessment → optimization strategy → implementation roadmap
- **Guideline Integration**: Properly extracts timeframes, implementation phases, milestones
- **Collaborative Guidance**: Provides next planning steps and local validation questions
- **Expert Positioning**: Presents as collaborative advisor, not directive authority

RED FLAGS (automatic score ≤2):
- Wrong methodology for query type (planning advice for factual query, or pure data for actionable query)
- Generic advice that ignores municipal context
- Missing key methodology components for query type

3. MUNICIPAL RELEVANCE (1-5):
CRITICAL CHECKS - This is where most responses fail:
- **Specificity**: Are insights/recommendations specific to THIS municipality, not generic?
- **Scale Appropriateness**: Are suggestions feasible at municipal scale and authority?
- **Query Alignment**: Does it directly address the specific question asked?
- **Local Context**: Does it consider municipal-specific constraints and opportunities?
- **Implementation Feasibility**: Are next steps actionable by municipal decision-makers?

RED FLAGS (automatic score ≤2):
- Generic recommendations applicable to any municipality
- Suggesting actions outside municipal authority
- Not answering the specific question asked
- Ignoring obvious municipal constraints or context

4. TECHNICAL COMPLIANCE (1-5):
FORMAT REQUIREMENTS:
- **Language Consistency**: Response in correct language throughout
- **Header Structure**: Uses only H3 (###) and H4 (####) headers
- **Citation Format**: Official sources cited as **Source Name** format
- **Structure Quality**: Appropriate use of tables, bullets, formatting

CONTENT REQUIREMENTS:
- **Source Accuracy**: Citations actually support the claims made
- **Professional Presentation**: Hides internal tool names, maintains expert positioning
- **Completeness**: Includes required conclusion sections (next steps, recommendations)

RED FLAGS (automatic score ≤2):
- Incorrect citation format or false citations
- Wrong header levels or poor formatting
- Missing required sections for query type
- Language inconsistencies

SCORING PHILOSOPHY:
- Score 1: Fundamental misunderstanding, major errors, or advice that is actively misleading or harmful.
- Score 2: Wrong methodology, significant mistakes, or advice that is not actionable or fails basic requirements.
- Score 3: Basic competence but significant limitations, generic advice, or methodology gaps
- Score 4: Good quality with minor issues, genuinely useful for municipal planning, correct methodology
- Score 5: Exceptional - accurate, specific, actionable, demonstrates deep understanding, perfect methodology alignment

BE ESPECIALLY CRITICAL OF:
- Responses that include irrelevant data just because it was retrieved
- Professional-sounding advice that's actually generic or inappropriate
- Methodology mismatches (planning advice for data queries, or vice versa)
- Mathematical errors disguised by confident presentation
- Claims not supported by actual data or guidelines

SCORING FORMAT:
{scoring_format}

Remember: Your job is to catch responses that sound authoritative but provide poor energy planning guidance. Be especially harsh on data misuse and methodology misalignment.
""")
```

Prompt 15 - Benchmark, system prompt

Table of Acronyms

AI :	Artificial Intelligence
API :	Application Programming Interface
FSM :	Finite-State Machine
LLM :	Large Language Model
MLLM :	Multimodal Large Language Model
PDF :	Portable Document Format
RAG :	Retrieval-Augmented Generation
RLM :	Reasoning Language Model
SSE :	Server-Sent Events
WMTS :	Web Map Tile Service

Equations

TODO: CHECKER les maths encore une fois TODO: citer la source wikipedia???

$$\overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \text{ where } \overline{x} \text{ is the sampled mean, } N \text{ the number of samples and } x_i \text{ the } i_{\mathrm{th}} \text{ sample tile.} \quad (1)$$

$$s = \sqrt{\frac{\sum_{i=1}^{N(x_i - \overline{x})\ 2}}{N-1}} \text{ where } s \text{ is the sample standard deviation of a single tile.} \quad (2)$$

$$\text{Confidence interval for a single tile} = \overline{x} \pm t_{\frac{\alpha}{2},N-1} \cdot \frac{s}{\sqrt{N}}$$
where $t_{\frac{\alpha}{2},N-1}$ is the critical value from the T-distribution for confidence level $1 - \alpha$ and $N - 1$ degrees of freedom. (3)

$$\text{Cosine similarity} := \cos(\theta) = \frac{\mathrm{A} \cdot \mathrm{B}}{\|\mathrm{A}\|\|\mathrm{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} {A_i}^2} \cdot \sqrt{\sum_{i=1}^{n} {B_i}^2}}$$
where θ is the angle between A and B, two n-dimensional vectors and A_i, B_i the i_{th} components of vectors A and B. (4)

$$\text{quartile coefficient of dispersion} := \frac{\frac{1}{2}\mathrm{IQR}}{\frac{Q_3+Q_1}{2}} = \frac{\frac{1}{2}(Q_3 - Q_1)}{\frac{Q_3+Q_1}{2}} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$
where IQR is the interquartile range and Q_1 and Q_3 the first and third quartiles, respectively. (5)

List of Figures